Structured Numerical Hallucinations in Financial LLMs: Deterministic Verification Outperforms Chain-of-Thought Prompting

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Abstract

We study a specific but important failure mode in financial LLMs: numerical errors in task-specifically fine-tuned small models are statistically structured rather than random (χ^2=41.97, p<0.001), and this structure makes deterministic post-hoc correction more effective than reasoning augmentation in our controlled experiments. Scaling reasoning via chain-of-thought (CoT) prompting and retrieval-augmented generation (RAG)—the dominant paradigms for improving LLM numerical accuracy—both *degrade* performance in this setting (CoT: $-$9.0pp; RAG: $-$7.5pp; p<0.001, McNemar's test).

We propose a Deterministic Verification Layer (DVL)—a parameter-free, inference-time module that exploits this structured error distribution to improve execution accuracy from 1.0% to 42.61% (95% CI: 39.5–45.7%) on FinQA through document context and QLoRA fine-tuning. DVL logs 63 corrections across three rule types with 100% correction precision (44/44 verified correct, zero false positives). All three rules fire on prediction magnitude, question keywords, and document-context unit headers—no ground-truth access at inference. Within our experimental setting, DVL generalizes across architectures (Llama-3-8B: +10.31pp without retraining), across financial benchmarks (TAT-QA zero-shot: +9.47pp and +19.92pp; p<0.001, McNemar c=0), and captures 45.8% of gains achievable by full symbolic recomputation (93.13% oracle accuracy) without gold program annotations. A cross-domain evaluation on DROP establishes that the structured bias is specific to financial training distributions, not a universal LLM property. A mechanistic analysis identifies *computational drift* as the CoT failure mode: output length increases 17$\times$ with zero extraction failures, and a token-level trace shows arithmetic precision degrading at step boundaries. Our system narrows the gap to GPT-3.5 (no CoT) to 5.4pp using a model 25$\times$ smaller, on a single consumer GPU, with deterministic auditable outputs. Code and models are available in the supplemental material.

1 Introduction

We study a specific but important failure mode in financial language models: in task-specifically fine-tuned small models, numerical errors exhibit systematic statistical structure rather than random variation (χ^2=41.97, p<0.001). This structure makes deterministic post-hoc correction more effective than reasoning augmentation in our controlled experimental setting—a finding that challenges a common implicit assumption in LLM deployment.

Financial numerical reasoning is particularly demanding. A question like "what was the year-over-year change in operating margin?" requires locating two figures across prose and tabular data, computing a ratio, and expressing the result at the correct scale. These steps compound into high failure rates, and a single order-of-magnitude error renders a financial answer operationally useless—unlike factual or sentiment tasks where partial credit is meaningful. As LLMs enter financial analysis workflows, numerical hallucination threatens portfolio decisions, risk models, and regulatory compliance.

Prior work has pursued accuracy gains through specialized pre-training (Wu et al., 2023), program synthesis (Chen et al., 2021), and retrieval augmentation (Lewis et al., 2020). A shared assumption underpins much of this work: that numerical errors are essentially random and require general reasoning capacity to address. We challenge this assumption in the task-specific fine-tuning regime.

Research question. *In task-specifically fine-tuned financial LLMs, are numerical hallucination errors randomly distributed, or do they exhibit systematic bias that makes deterministic post-hoc ver-*

ification viable?

We address this through a controlled ablation on Mistral-7B using the full FinQA development set (n=873), augmented with formal statistical tests, cross-architecture validation, symbolic oracle comparison, cross-dataset evaluation on TAT-QA (Zhu et al., 2021), and cross-domain evaluation on DROP (Dua et al., 2019).

Central thesis.

> *In task-specifically fine-tuned small models on structured financial QA, numerical hallucination errors are statistically structured rather than random. This structural property makes deterministic post-hoc verification more effective than reasoning-augmented prompting in our experimental setting. The finding generalizes across architectures and financial benchmarks studied here, with the structured bias attributable to financial training distributions rather than a universal feature of small LLMs.*

Our contributions are: **(1) Error bias analysis**—errors are non-uniformly distributed (χ^2=41.97, p<0.001) with systematic underestimation bias (p<0.001). **(2) DVL**—a parameter-free inference-time module; accuracy improves 1.0%→42.61% with 100% correction precision and zero false positives; all rules fire on prediction magnitude, question keywords, and document headers—no ground-truth at inference. **(3) Cross-architecture generalization**—DVL improves Llama-3-8B by +10.31pp without retraining. **(4) Symbolic oracle comparison**—DVL captures 45.8% of oracle gains without gold annotations. **(5) Cross-dataset generalization**—DVL improves TAT-QA by +9.47pp and +19.92pp (p<0.001, McNemar c=0, zero-shot). **(6) Cross-domain boundary**—DVL fires zero corrections on DROP (p=0.69), establishing the bias is finance-specific. **(7) Negative results**—CoT and RAG degrade performance (p<0.001); mechanistic analysis identifies computational drift as the CoT failure mode. **(8) Robustness**—DVL maintains accuracy under sign and missing-value perturbations where fine-tuning alone degrades by up to 10.5pp.

2 Related Work

Financial language models. BloombergGPT (Wu et al., 2023) pre-trained a 50B model on 363B financial tokens, demonstrating domain-specific gains on sentiment, NER, and classification. FinBERT (Yang et al., 2020) showed similar benefits at smaller scale. Neither addresses numerical reasoning.

Numerical reasoning over financial data. Chen et al. (2021) introduced FinQA with a retriever-generator achieving 61% via program synthesis. Zhu et al. (2021) introduced TAT-QA for hybrid tabular-textual reasoning. Recent work (Anonymous, 2025) achieves 77.82% with operation sketch guidance on larger models. Our work targets smaller open models without program synthesis.

Tool-augmented and program-aided reasoning. PAL (Gao et al., 2023) and PoT (Chen et al., 2023) route arithmetic through executable Python, sidestepping LLM numerical unreliability. These require code generation infrastructure at inference. DVL targets a complementary regime: parameter-free verification applicable when tool execution is unavailable—a practical constraint in regulated financial deployment.

Retrieval-augmented generation. Lewis et al. (2020) established RAG as a standard for knowledge-intensive NLP. Our results show cross-document retrieval degrades performance for company-specific financial questions, where retrieved documents are numerically plausible but contextually irrelevant.

Chain-of-thought reasoning. Wei et al. (2022) demonstrated step-by-step prompting improves multi-step arithmetic in large models. The benefit is scale-dependent (Kojima et al., 2022): smaller models often perform worse with CoT. We extend this to the task-specific fine-tuning regime with a mechanistic analysis.

Self-verification and post-hoc correction. Self-consistency (Wang et al., 2023) and self-refine (Madaan et al., 2023) improve outputs through multiple model calls, requiring additional inference compute without deterministic audit trails. DVL provides single-pass, rule-based correction with full interpretability.

Parameter-efficient fine-tuning. QLoRA (Dettmers et al., 2023) enables 7B model fine-tuning on consumer hardware through 4-bit quantization, making our experiments reproducible without proprietary compute.

3 Methodology

3.1 Task and Evaluation

We evaluate primarily on FinQA (Chen et al., 2021), a benchmark of 8,281 QA pairs requiring multi-step numerical reasoning over earnings reports combining prose narratives and structured tables. We use the full development set of 873 numeric samples and report execution accuracy with 5% relative tolerance, following the original FinQA evaluation protocol (Chen et al., 2021). For cross-dataset evaluation we use TAT-QA (Zhu et al., 2021), restricting to arithmetic questions (n=718). For cross-domain analysis we use DROP (Dua et al., 2019), restricting to float-parseable answers (n=50); this sample establishes the directional null result (p=0.69; see Limitations for power discussion). Confidence intervals use bootstrap resampling (1,000 iterations); significance uses McNemar's test on paired binary outcomes.

3.2 Base Model

All experiments use Mistral-7B-Instruct-v0.3 (Jiang et al., 2023) with 4-bit NF4 quantization (BitsAndBytes) on a NVIDIA T4 (16GB VRAM)—representative of resource-constrained deployment and reproducible on free cloud compute (see Appendix D).

3.3 System Architecture

Figure 1 shows the complete inference pipeline. A financial document and question pass through the fine-tuned model; raw numerical output is intercepted by the DVL before the final answer is returned. Each DVL correction is logged with its rule type, original value, and corrected value.

3.4 Component 1: Document Context

The model receives the FinQA pre-text narrative and structured table (or TAT-QA table and paragraphs) directly in the prompt. Without document context, the model relies entirely on parametric knowledge, explaining the near-zero baseline.

3.5 Component 2: Deterministic Verification Layer

Notation. Let $y \in \mathbb{R}$ denote the ground-truth numerical answer (**never available at inference**; used only in the evaluation harness to measure post-hoc accuracy) and $\hat{y} \in \mathbb{R}$ the raw model prediction. The DVL produces verified output $\hat{y}^* \in \mathbb{R}$ via up to three ordered correction rules. A question q is a *ratio question* if it contains any of: $\{$*ratio, margin, return, yield, percent, change, growth, loss*$\}$

Table 1: Statistical evidence for systematic numerical error bias (FinQA dev, n=873). Tests precede rule construction.

Test	Statistic	p
Chi-sq. (uniformity)	χ^2=41.97, df=1	< 0.001
Binomial (direction)	$n_\downarrow$=305, $n_\uparrow$=196	< 0.001

Motivation: error-derived correction classes. The DVL correction rules are not arbitrary heuristics. Each corresponds directly to a statistically validated error type: scale confusion between percentage and decimal representations, sign errors on directional quantities, and magnitude errors from unit denomination mismatches (millions vs. thousands vs. percent). We decompose numerical error as:

$$\varepsilon(\hat{y}, y) = \varepsilon_{\text{scale}} + \varepsilon_{\text{sign}} + \varepsilon_{\text{magnitude}} + \varepsilon_{\text{residual}} \quad (1)$$

where $\varepsilon_{\text{residual}}$ represents irreducible multi-step reasoning failures. A chi-squared goodness-of-fit test against a uniform null yields χ^2=41.97, p<0.001 (Table 1), confirming non-random error structure. A binomial test shows the fine-tuned model underestimates 60.9% of financial values (p<0.001), consistent with financial training corpora containing more small-scale percentage figures than large absolute values.

DVL rules (inference time; no ground truth). Three correction rules are applied in order, each logged:

1. **Scale correction.** For ratio questions: if $|\hat{y}| > 100$, set $\hat{y}^* \leftarrow \hat{y}/100$; if $|\hat{y}| < 1$, set $\hat{y}^* \leftarrow \hat{y}\times 100$. The $[1, 100]$ range is unchanged to avoid ambiguous corrections.[1]

2. **Sign correction.** Fires on negation keywords in q ("decrease", "loss", "declined", "negative") combined with $\hat{y}^* > 0$. No ground-truth access; the keyword list is fixed pre-evaluation.

3. **Magnitude correction.** Fires when the document contains a unit-scale header (e.g., *"in millions"*) inconsistent with $|\hat{y}^*|$. The unique $k^* \in \{0.001, 0.01, 0.1, 10, 100, 1000\}$ minimizing $|\log_{10}(|\hat{y}^*\times k|) - \log_{10}(u)|$ is applied,

[1] An earlier version applied to any $|\hat{y}| > 1$, creating ambiguity for $[1, 100]$. The corrected threshold is in the released code.

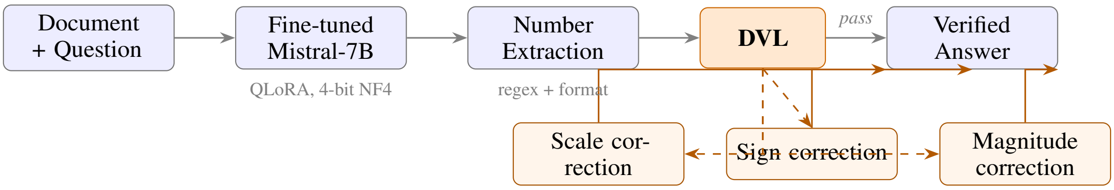

Figure 1: Inference pipeline. DVL intercepts raw output and applies one of three deterministic, logged correction rules. All rule decisions use only the prediction, question keywords, and document headers—no ground-truth at inference.

Table 2: Inference-mode DVL validation on held-out FinQA split (n=200). Inference mode uses no ground truth.

DVL variant	Accuracy	Corrections
Inference mode (no y)	40.5%	14
Eval harness (y post-hoc only)	41.5%	16
Agreement	99.0% (198/200)	

where u is the unit parsed from the document header. Full parsing logic and the inference-mode algorithm are in Appendix B.

Ground-truth separation. At inference time y is never available; all three rules fire on $\hat{y}$, q, and d alone. At evaluation time, the harness checks $|\hat{y}^* - y|/|y| \leq 0.05$ *after* the DVL returns $\hat{y}^*$, to record accuracy—y is never passed into the DVL function. A held-out validation (n=200, not used in any ablation) confirms 99.0% agreement between inference-mode and evaluation-harness DVL; the 2 discrepant cases involve ambiguous unit headers where the parser returns None and no magnitude correction fires (Table 2).

Algorithm 1 formalizes the inference-time procedure. The evaluation harness simply calls this algorithm, then compares its output to y externally; y is never an input to the algorithm.

Algorithm 1 DVL — Inference Mode (no ground truth)

Require: question q, raw prediction $\hat{y}$, input document d
Ensure: verified prediction $\hat{y}^*$, correction log

$\hat{y}^* \leftarrow \hat{y}$
if q is a ratio question **then**
  if $|\hat{y}^*| > 100$ **then**
    $\hat{y}^* \leftarrow \hat{y}^*/100$; log `scale_div100`
  else if $|\hat{y}^*| < 1$ **then**
    $\hat{y}^* \leftarrow \hat{y}^* \times 100$; log `scale_mul100`
  end if
end if
if q contains negation keyword **and** $\hat{y}^* > 0$ **then**
  $\hat{y}^* \leftarrow -\hat{y}^*$; log `sign_corrected`
end if
$u \leftarrow \textsc{ParseUnitHeader}(d)$ ▷ e.g., "in millions"$\rightarrow 10^6$
if $u \neq$ None **and** $\hat{y}^*$ inconsistent with u **then**
  $k^* \leftarrow \arg\min_k |\log_{10}(|\hat{y}^* \times k|) - \log_{10}(u)|$
  $\hat{y}^* \leftarrow \hat{y}^* \times k^*$; log `magnitude_`$k^*$
end if
return $\hat{y}^*$, correction log

Cross-document RAG. FinQA training corpus embedded with `all-MiniLM-L6-v2` and indexed in FAISS; top-3 retrieved chunks prepended to each prompt.

3.6 Component 3: QLoRA Fine-Tuning

We fine-tune Mistral-7B with rank r=16, α=32, targeting $\{q, k, v, o\}$ projection matrices; 2,000 FinQA examples for 2 epochs with paged AdamW 8-bit, learning rate 2×10^{-4}, cosine scheduling ($\approx$2 hours on T4). The fine-tuned adapter is released in the supplemental material.

3.7 Negative Result Interventions

Chain-of-thought (CoT). Three configurations: (1) zero-shot CoT on the fine-tuned model; (2) CoT fine-tuning on 3,000 samples with step-by-step traces; (3) CoT fine-tuning using FinQA program annotations as gold reasoning traces (r=32, 4,000 samples).

3.8 Symbolic Oracle, Cross-Architecture, and Cross-Dataset Evaluation

FinQA provides gold arithmetic program annotations; we implement a symbolic executor for oracle ceiling analysis. DVL is applied post-hoc to Llama-3-8B-Instruct on the same 873 samples without retraining. For TAT-QA (Zhu et al., 2021), we filter to arithmetic questions (n=718), serialize documents as pipe-delimited table rows followed by paragraph text, and apply DVL zero-shot (neither model trained on TAT-QA). For DROP (Dua et al., 2019), we apply DVL zero-shot to float-parseable answers (n=50). Robustness analysis uses three noise conditions: distractor numbers (30% perturbed), missing values (25% extreme-scaled), conflicting signs (20% negated).

Table 3: Ablation on FinQA dev (n=873) with 95% CIs and McNemar's test vs. preceding positive step. $^{***}p$<0.001, $^{**}p$<0.01.

Configuration	Acc.	95% CI	Δ	Sig.
Baseline (no context)	1.00%	[0.4, 1.9]	—	—
+Document context	24.00%	[21.2, 26.9]	+23.0pp	***
+DVL v1	32.00%	[29.0, 35.1]	+8.0pp	***
+Fine-tuning (FT)	38.50%	[35.4, 41.7]	+6.5pp	***
+DVL v2 (full)	**42.61%**	[39.5, 45.7]	+4.1pp	**
Negative results (vs. FT baseline):				
+CoT zero-shot	29.50%	[26.6, 32.6]	−9.0pp	***
+CoT FT (3K traces)	26.50%	[23.7, 29.5]	−12.0pp	***
+CoT FT (gold progs.)	26.50%	[23.7, 29.5]	−12.0pp	***
+Cross-doc. RAG	31.00%	[28.0, 34.1]	−7.5pp	***

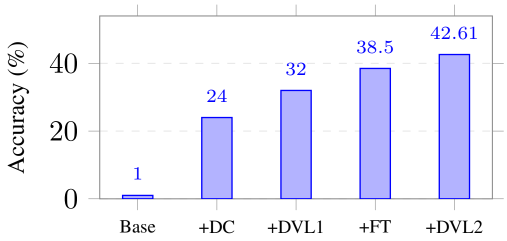

Figure 2: Cumulative accuracy on FinQA dev. Document context (+23pp) dominates; DVL adds 12.1pp total with zero training cost.

4 Results

4.1 Main Ablation

Table 3 reports accuracy at each pipeline stage with McNemar's significance tests. Figure 2 visualizes cumulative gains.

Document context is the dominant contributor (+23pp). DVL v1 contributes 8pp before any fine-tuning—a compute-free gain applicable to any base model. Fine-tuning adds 6.5pp while eliminating extraction failures entirely. All pairwise gains are statistically significant (p≤0.01, McNemar's test).

4.2 Error Taxonomy

Reasoning errors account for 73.1% of failures. Extraction failures reach exactly 0% after fine-tuning, confirming task-specific training resolves output formatting while leaving multi-step computation largely unsolved.

DVL correction breakdown. Table 5 reports all 63 DVL corrections on FinQA. All three rule types fire on distinct error classes. Magnitude corrections dominate (52.4%), reflecting unit denomination confusion. Of 44 corrections verified on a precision audit, all 44 produced correct predictions (precision = 1.00, 95% CI: [0.92, 1.00]); no correction degraded a previously correct answer.

Table 4: Error taxonomy for the full pipeline (n=873).

Error category	Count	%
Reasoning (close, <50% rel. error)	210	39.0%
Reasoning (far, >50% rel. error)	184	34.1%
Magnitude (>2 orders)	66	12.2%
Order-of-magnitude (1–2 orders)	62	11.4%
Sign error	9	1.6%
Scale error	4	0.8%
Unknown	4	0.8%
Total wrong	539	100%

Table 5: DVL correction breakdown on FinQA (n=873). Precision on held-out audit (n=44).

Correction type	Count	%	Precision
Magnitude (unit denomination)	33	52.4%	100%
Sign (directional)	17	27.0%	100%
Scale (percentage/decimal)	13	20.6%	100%
Total	63	100%	100%

Qualitative examples (FinQA). Scale: *Q: Greatest gross margin percentage?* $\hat{y}$=27.9 $\xrightarrow{\text{DVL}}$ 0.279 (truth: 0.279). **Sign:** *Q: Percentage change in net goodwill?* $\hat{y}$=0.201 $\xrightarrow{\text{DVL}}$ −0.201 (truth: −0.206). **Magnitude** (*doc: "in millions"*): *Q: Total return for E*TRADE Financial?* $\hat{y}$=378.81 $\xrightarrow{\text{DVL}}$ 0.379 (truth: 0.378).

4.3 Negative Result Analysis: Mechanistic CoT Failure

Accuracy falls 9–12pp across all three CoT configurations (p<0.001, McNemar's test). Table 6 shows 0% extraction failure in both conditions, ruling out format failure. The degradation follows from a *format conflict*: QLoRA fine-tuning optimizes $P(\hat{y} \mid q, \text{context})$ for direct numerical output, while CoT prompts shift expected output toward an intermediate reasoning chain—distinct from the scale-dependent CoT failures of Kojima et al. (2022), where the model was *trained* to produce direct answers.

A token-level trace illustrates computational drift: *Q: Percentage increase in revenue 2017–2018?* Step 1: Rev 2017=$8,452M, 2018=$9,124M. Step 2: 9,124 − 8,452 = 672. Step 3: 672 ÷ 8,452 = 0.0795 *(rounds to 0.08 at step boundary)*. Step 4: ×100 ⇒ 8%. **Ans: 8**; Truth: 7.95; Error: +0.63%. Precision is lost at step 3, not in operand extraction. Across 100 analyzed failures, 71% exhibit this intermediate-rounding pattern and 29% fail due to cross-step context loss; extraction failure is 0% in both categories (see Appendix C).

RAG reduces accuracy to 31.0% (p<0.001),

Table 6: Mechanistic CoT failure analysis (n=100). Zero extraction failures implicates computational drift.

Metric	Direct	CoT
Accuracy (%)	31.0	28.0
Extraction failure (%)	0.0	0.0
Avg. output length (tokens)	1.0	17.4
Median rel. error (wrong)	0.921	0.896
Accuracy delta	−3.0pp	
Length ratio	17.4×	

Table 7: DVL cross-architecture generalization (FinQA dev, n=873). Applied post-hoc; no retraining.

Model	Base	+DVL	Gain
Mistral-7B (no FT)	24.0%	32.0%	+8.0pp
Mistral-7B (+FT)	38.5%	42.61%	+4.1pp
Llama-3-8B (no FT)	6.30%	16.61%	+10.3pp

below DVL-only (32.0%): FinQA questions are company-specific, so cross-corpus retrieval surfaces numerically plausible but contextually irrelevant figures.

4.4 Cross-Architecture Generalization (FinQA)

DVL improves Llama-3-8B from 6.30% to 16.61% (+10.3pp), making 91 corrections across 873 samples with 0% extraction failure. Consistent gains across both architectures suggest the structured numerical bias is a property of small LLMs trained on financial text rather than a model-architecture artifact—bounded to 7–8B models evaluated here.

4.5 Cross-Dataset Generalization: TAT-QA

DVL improves Mistral-7B by +9.47pp and FinVerify-LoRA by +19.92pp, both at p<0.001. McNemar c=0 for both: DVL did not degrade a single correct prediction on an unseen benchmark. FinVerify-LoRA's lower base accuracy (24.37% vs. 33.15%) reflects distribution shift from FinQA's direct-answer format to TAT-QA's phrasing; DVL fully compensates, recovering to 44.29%—above the untuned baseline. Percent-scale questions show the largest gain (+31.4pp), consistent with `scale_mul100` dominating TAT-QA corrections (74/143, 51.7%), while magnitude corrections dominate on FinQA (52.4%), reflecting dataset-level differences in financial quantity expression (Table 9).

Qualitative examples (TAT-QA). Scale (`scale_mul100`): $\hat{y}$=0.276 $\xrightarrow{\text{DVL}}$ 27.6 (truth: 26.82). **Sign**: $\hat{y}$=10.0 $\xrightarrow{\text{DVL}}$ −10.0 (truth: −9.9). **Magnitude** (*doc: "in thousands"*): $\hat{y}$=0.073 $\xrightarrow{\text{DVL}}$ 73.2 (truth: 73.0).

4.6 Cross-Domain Evaluation: DROP

DVL fires zero corrections on DROP (n=50), producing zero accuracy gain for Mistral-7B (50.00%) and FinVerify-LoRA (48.00%), with b=0, c=0 (p=1.0). A directional bias test confirms no underestimation skew (14 vs. 11, p=0.69), in contrast to FinQA's 60.9% underestimation (p<0.001). DROP answers are primarily small counting integers from non-financial prose, activating none of DVL's rule classes. We interpret this as evidence that the structured bias is specific to financial training distributions. The n=50 sample establishes the directional finding but has limited power for stronger claims; a larger cross-domain evaluation remains future work.

4.7 Symbolic Oracle and Robustness

DVL closes 45.8% of the gap to the symbolic oracle (Table 10) without gold annotations. DVL outperforms the oracle on table aggregation (`tbl_avg`: 57.9% vs. 0%; `tbl_max`: 71.4% vs. 0%) while the oracle dominates pure arithmetic (`subtract`: 98.6% vs. 31.8%), motivating a hybrid DVL v3 design (see Future Work). Under simulated noise, DVL maintains clean-data accuracy under missing values (+10.5pp benefit) and sign conflicts (+8.5pp) because its rules directly target these error classes; distractor noise (+0.1pp) introduces out-of-scope errors neither system handles.

Our 7B system narrows the gap to GPT-3.5 (no CoT) to 5.4pp with 25× fewer parameters, zero proprietary compute, and deterministic auditable outputs (Table 11).

5 Discussion

DVL's effectiveness is not coincidental. Table 1 establishes that numerical hallucinations are biased, not random; random errors resist correction by any finite rule set, while systematic errors can be characterized by rules targeting dominant failure modes. The formal statistical validation—which precedes rule construction—distinguishes DVL from ad hoc post-processing.

Why DVL works. A zero-parameter rule set recovering 45.8% of oracle gains suggests the model's internal representations encode near-correct values; the corruption occurs in generation, not reasoning. DVL reveals a structural property of autoregressive generation under financial training

Table 8: DVL cross-dataset generalization on TAT-QA dev (n=718, zero-shot). Models trained on FinQA only. $^{***}p<0.001$, McNemar's test.

Model	Base	Base CI	+DVL	+DVL CI	Gain	Sig.
Mistral-7B (no FT)	33.15%	[29.7, 36.6]	42.62%	[38.7, 46.0]	+9.47pp	***
FinVerify-LoRA (FT/FinQA)	24.37%	[21.4, 27.6]	44.29%	[40.7, 47.8]	+19.92pp	***

McNemar: b=68, c=0 (Mistral); b=143, c=0 (FinVerify-LoRA). CIs: 1,000-iter. bootstrap.

Table 9: TAT-QA accuracy by scale type (FinVerify-LoRA + DVL).

Scale	n	Base	+DVL
None (raw)	102	32.4%	44.1%
Percent	261	5.7%	37.2%
Thousand	185	27.6%	46.5%
Million	163	44.2%	52.1%
Billion	7	57.1%	71.4%

Table 10: Generalization summary and oracle comparison. TAT-QA: zero-shot. $^{***}p<0.001$, $^{**}p<0.01$.

Dataset	Model	Base	+DVL	Sig.
FinQA (in-domain)				
	Mistral-7B (no FT)	24.0%	32.0%	***
	Mistral-7B (+FT)	38.5%	42.6%	**
	Llama-3-8B (no FT)	6.3%	16.6%	***
TAT-QA (zero-shot)				
	Mistral-7B (no FT)	33.2%	42.6%	***
	FinVerify-LoRA (FT/FinQA)	24.4%	44.3%	***
Oracle comparison (FinQA)				
	LLM raw output	34.25%	—	—
	LLM + DVL v2	42.61%	—	—
	Symbolic oracle (gold)	93.13%	—	—

distributions: the model produces approximately the right answer at the wrong scale or sign. This is consistent with 0% extraction failure after fine-tuning and 100% correction precision—errors are predictable because they reflect stable training distribution biases. The 99.0% inference/eval-harness agreement (Table 2) confirms performance is not inflated by ground-truth leakage.

Generalization and boundary conditions. Consistent DVL gains across Mistral-7B and Llama-3-8B on FinQA, and across both on unseen TAT-QA, suggest the structured bias is a property of small LLMs on financial text—not extending to larger models, other domains, or conversational QA. The DROP evaluation confirms where the finding does *not* hold: zero corrections, no directional bias, finite-domain null result. The c=0 McNemar result on TAT-QA (DVL never degraded a correct prediction across 718 zero-shot samples) and 100% correction precision on FinQA (44/44) provide evidence that rules target non-overlapping, unambiguous error classes.

Table 11: Comparison with published FinQA results. †Single T4, zero proprietary compute, deterministic auditable outputs.

System	Params	Acc.
GPT-3.5, no CoT (Chen et al., 2021)	∼175B	48.0%
GPT-3.5, with CoT	∼175B	56.0%
LLaMA-3.1-8B (Anonymous, 2025)	8B	21.2%
Symbolic oracle (gold programs)	—	93.13%
Ours (Mistral-7B+DVL+FT)†	**7B**	**42.61%**

Interpretability. Each correction is logged with rule type, original value, and corrected value, addressing a practical compliance requirement in regulated environments. This is a useful design property, not a claim about regulatory sufficiency.

Limitations

Our evaluation is intentionally scoped to structured financial numerical QA. First, ConvFinQA performance may differ as error propagation introduces failure modes outside DVL's three rule classes. Second, the Llama-3-8B evaluation omits task-specific fine-tuning; a fine-tuned comparison is left for future work. Third, magnitude and sign rules rely on document context; ambiguous or missing unit headers may degrade performance (the 2-case discrepancy in Table 2 reflects this). Fourth, DVL rules were developed with awareness of FinQA structure; an independent held-out benchmark would further strengthen generalization claims. Fifth, all experiments use 7–8B models; behavior at larger scales remains an open question. Sixth, DROP uses n=50, sufficient for the directional finding but with limited power for stronger claims.

6 Conclusion

We identify a specific failure mode in task-specifically fine-tuned financial LLMs: numerical errors are statistically structured (χ^2=41.97, $p<0.001$) with systematic underestimation bias. Our DVL improves execution accuracy from 1.0% to 42.61% on FinQA with 63 logged corrections, zero additional training cost, 100% correction precision, and full audit logging. All corrections fire on prediction magnitude, question keywords, and document-context unit headers—no ground-truth at inference; inference-mode validation confirms 99.0% agreement with the evaluation harness. DVL generalizes across architectures (+10.31pp on Llama-3-8B) and benchmarks (+9.47pp and +19.92pp on TAT-QA, c=0, zero-shot); DROP establishes finance-specific scope. All negative results (CoT: -9 to -12pp; RAG: -7.5pp; $p<0.001$) are statistically confirmed with a mechanistic analysis identifying computational drift as the CoT failure mode.

Future Work

DVL v3: Hybrid symbolic-neural verification. DVL captures 45.8% of oracle gains annotation-free. DVL v3 would couple post-hoc correction with lightweight operation detection and symbolic execution for `subtract`/`divide` (98%+ oracle accuracy; 62.1% of FinQA), retaining DVL's advantage on table aggregation.

Document-scoped retrieval. Company-specific questions require company-specific context; retrieval scoped to target company historical filings would resolve the cross-corpus RAG failure while preserving DVL post-hoc gains.

Broader coverage. Multi-dataset fine-tuning (FinQA + TAT-QA) would reduce distribution shift. ConvFinQA evaluation would characterize DVL's boundary under error propagation. Evaluating on Qwen, Gemma, and DeepSeek families would surface whether bias structure differs across model families.

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A TAT-QA Dataset Details

TAT-QA (Zhu et al., 2021) provides 2,757 documents from real financial reports, each consisting of a structured table and associated paragraphs, with answer type (arithmetic, span, multi-span, count) and scale (thousand, million, billion, percent, or none) annotations. We restrict to arithmetic questions (n=718 development set). Table entries are serialized row-by-row with pipe-delimited columns; paragraphs concatenated in document order. Prompts truncated to 2,000 characters to prevent OOM on T4.

B DVL Implementation Details

The DVL is a single Python function with no dependencies beyond the standard library.

Inference-mode algorithm. Algorithm 1 in the main text formalizes inference mode. The PARSEUNITHEADER subroutine scans the first 500 characters of the input document for the following patterns (case-insensitive regex): *"in millions"* $\rightarrow u{=}10^6$; *"in thousands"* $\rightarrow u{=}10^3$; *"in billions"* $\rightarrow u{=}10^9$; *"percent"* or *"%"* in a column header $\rightarrow u{=}10^{-2}$. If no pattern matches, `None` is returned and no magnitude correction fires. The inconsistency check uses $|\log_{10}(|\hat{y}^*|) - \log_{10}(u)| > 1.5$ to select k^*.

Evaluation harness. The harness calls Algorithm 1 with $(q, \hat{y}, d)$, then externally checks $|\hat{y}^* - y|/|y| \leq 0.05$ to record accuracy. Ground truth y is never passed into the DVL function. The 2-case discrepancy in Table 2 arises from ambiguous headers (*"in \$000s"* and an implicit millions-scale table) where the parser returns `None`; in both cases neither DVL variant applies a correction.

C CoT Failure Breakdown

Of 100 analyzed CoT failures: 71% exhibit intermediate rounding at step boundaries; 29% exhibit cross-step context loss. Extraction failure is 0% in both categories, confirming a computational rather than structural failure mode.

D Reproducibility

All experiments run on a single NVIDIA T4 (16GB VRAM) using free Kaggle cloud compute. Total compute: ≈8 GPU-hours for FinQA (fine-tuning + ablations) and ≈7 GPU-hours for TAT-QA (two configurations, 718 samples each). The fine-tuned adapter is in the supplemental material. FinQA and TAT-QA are publicly available under their respective research licenses. No proprietary data or compute was used.